








Advanced Time Series Analysis and Forecasting

Chapter 1: Forecast evaluation, scoring rules and combination



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **ALFRED**: ArchivaL Federal Reserve Economic Data
- **AO**: Atkeson–Ohanian (naive inflation forecast)
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARX**: AutoRegressive model with eXogenous variables
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CPI**: Consumer Price Index
- **CRPS**: Continuous Ranked Probability Score
- **CW**: Clark–West (test for nested models)
- **DM**: Diebold–Mariano (test)
- **EA**: Euro Area
- **ECB**: European Central Bank
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-N**: GARCH with Normal innovations
- **GDP**: Gross Domestic Product
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))

Acronyms used in this material (2/2)

- **ISE**: (Fraunhofer) Institute for Solar Energy Systems
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MAPE**: Mean Absolute Percentage Error
- **MASE**: Mean Absolute Scaled Error
- **MCS**: Model Confidence Set
- **MSE**: Mean Squared Error
- **MSPE**: Mean Squared Prediction Error
- **MW**: MegaWatt
- **MZ**: Mincer–Zarnowitz (regression)
- **OLS**: Ordinary Least Squares
- **PIT**: Probability Integral Transform
- **QLIKE**: Quasi-LIKElihood loss
- **RC**: White's Reality Check (for data snooping)
- **RMSE**: Root Mean Squared Error
- **RTDSM**: Real-Time Data Set for Macroeconomists (Philadelphia Fed)
- **RW**: Random Walk
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **SPF**: Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia)
- **TSA**: Time Series Analysis
- **UIP**: Uncovered Interest Parity
- **US**: United States
- **VAR**: Vector AutoRegression

Today's question and route

- **Question**: when is one forecast *really* better than another, and what do we gain by combining them?
  - "better" needs a loss or a score, a reference distribution of the comparison and an honest evaluation design
- **Route** of the chapter
  - point forecasts: which functional a loss function rewards (consistency, Bregman losses)
  - probabilistic forecasts: calibration, sharpness, PIT, proper scoring rules, elicitability
  - comparing forecasts: DM with HAC and small-sample corrections, GW, Clark–West, MZ, encompassing, RC, SPA, MCS
  - combination: Bates–Granger, the combination puzzle, density pools, the M4 and M5 lessons; real-time data
- We build on TSA, Chapter 0 and TSA, Chapter 4 (MAE, RMSE, MASE, rolling-origin evaluation, the basic DM test) and on Chapter 0 (HAC, bootstrap); Seminar 1 comes before this lecture

Learning outcomes

- Derive the optimal point forecast under squared, absolute, pinball and asymmetric losses, and recognise the losses consistent for the mean (Bregman)
- Assess the calibration of density forecasts with PIT histograms and tests, and rank them with proper scores (log score, CRPS, interval and energy scores)
- Explain elicitability and why VaR is elicitable while ES is elicitable only jointly with VaR
- Test equal predictive ability correctly: DM–HLN with HAC variance, GW, Clark–West for nested models, MZ, encompassing, SPA and the MCS
- Combine point and density forecasts and explain the forecast combination puzzle; evaluate forecasts with real-time data

Reading, data and tools

- Surveys: Petropoulos et al. (2022) (open access), Gneiting and Katzfuss (2014), Elliott and Timmermann (2008), Timmermann (2006), Wang et al. (2023)
  - textbooks: Hamilton (1994), Ch. 4 (forecasting); Kilian and Lütkepohl (2017); bridge: Huang and Petukhina (2022)
- Python Quantlets of this chapter: Quantlets/Ch_01
  - DM–HLN, GW, Clark–West, MZ, encompassing, Berkowitz, CRPS and the MCS written out in `numpy`; Hansen's SPA from the `arch` package
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
HICP inflation, Romania (y/y, monthly)	Eurostat (prc_hicp_minr)	DM, GW
CPI and unemployment, US (quarterly)	FRED (CPIAUCSL, UNRATE)	Atkeson–Ohanian
EUR/RON and 3-month rates RO, EA (monthly)	BNR reference rate; Eurostat (irt_st_m)	Clark–West, SPA
S&P 500, daily close	EODHD	PIT, scores
Electricity load, Romania (hourly)	Energy-Charts (Fraunhofer ISE), ENTSO-E data	pinball, MCS
Individual forecasts, US SPF	Federal Reserve Bank of Philadelphia	combination
US real GDP, all releases	Philadelphia Fed, RTDSM	real time

- All sources are public and need no account or key; daily and monthly data up to September 2026
- HICP: harmonised index of consumer prices; CPI: consumer price index; SPF: Survey of Professional Forecasters; RTDSM: Real-Time Data Set for Macroeconomists

Origins of forecast verification



Robert FitzRoy (1805–1865)

Portrait: Charles Hemus (c. 1910); public domain; Wikimedia Commons



Lewis Fry Richardson (1881–1953)

Photo: NOAA; public domain; Wikimedia Commons

- 1861: FitzRoy, head of the new Meteorological Office, publishes daily "forecasts" in The Times
  - he coined the word in its modern sense; the forecasts were often wrong and publicly mocked
- 1922: Richardson computes a weather forecast by numerical integration of the equations of motion, by hand
- 1950: Brier (1950) proposes a score for probability forecasts of rain; meteorology leads forecast verification for the next fifty years
  - economics adopts the same tools late: PIT Diebold, Gunther and Tay (1998), proper scores Gneiting and Raftery (2007), calibration Gneiting, Balabdaoui and Raftery (2007)
- Lesson: a forecast is only as useful as the way we evaluate it

Setting and notation

- Target Y_{t+h}, information set $\mathcal{F}_t$, horizon h; a **point forecast** $\hat{y}_{t+h|t}$ is $\mathcal{F}_t$-measurable
  - $\mathcal{F}_t$-measurable: computed only from information available at t
  - a **probabilistic forecast** is a predictive distribution $F_{t+h|t}$ (density, quantiles, interval, ensemble)
- Out-of-sample design: R observations to estimate, P forecasts to evaluate, $R + P + h - 1 = T$
  - schemes: **fixed** (estimate once), **rolling** (last R), **recursive** (all past); TSA, Chapter 4 called this rolling-origin evaluation
  - T: the total sample; the last forecast, made at $R + P - 1$, targets T; the scheme matters for inference: West (1996) versus Giacomini and White (2006), Section 4
- Known from TSA, Chapter 0: MAE, RMSE, MAPE and the scale-free MASE Hyndman and Koehler (2006)
  - $\mathrm{MAE} = P^{-1}\sum|e_t|$, $\mathrm{RMSE} = (P^{-1}\sum e_t^2)^{1/2}$, $\mathrm{MAPE} = 100P^{-1}\sum|e_t/y_t|$; MASE: the MAE divided by the in-sample MAE of the naive forecast
  - here the question is different: *which* loss, and what does it reward?

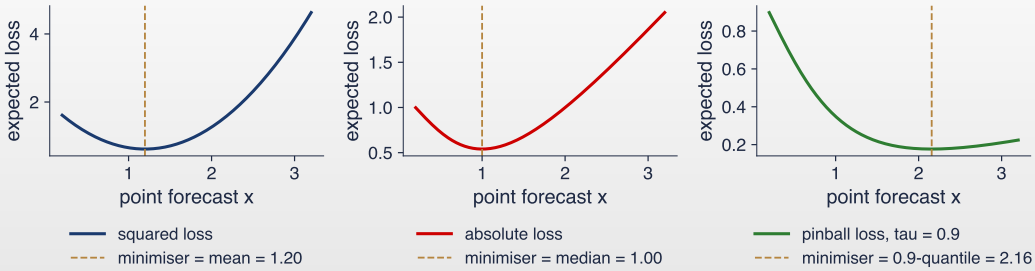
The optimal point forecast and consistency (1/2)

- Given a loss $L(x, y)$ and a predictive distribution F, the **optimal** (Bayes) forecast minimises the expected loss:
  - $x^* = \arg\min_x \mathbb{E}_F[L(x, Y)]$
  - x: the point forecast; y: the outcome; $L(x, y) \geq 0$: the penalty; $\mathbb{E}_F$: the expectation when $Y \sim F$
- A **functional** $\mathrm{T}(F)$ maps a distribution to a number: the mean, the median, a quantile
  - L is **consistent** for T if $\mathbb{E}_F L(\mathrm{T}(F), Y) \leq \mathbb{E}_F L(x, Y)$ for all x and F Gneiting (2011)
  - i.e. reporting $\mathrm{T}(F)$ is optimal whatever F is; **strictly** consistent: equality only at $x = \mathrm{T}(F)$

The optimal point forecast and consistency (2/2)

- Three classical cases, with the forecast error $e = y - x$; set the derivative of the expected loss to zero
  - squared e^2: $\frac{d}{dx}\mathbb{E}(Y-x)^2 = -2\mathbb{E}(Y-x) = 0 \Rightarrow x^* = \mathbb{E}Y$ (the mean)
  - absolute $|e|$: $\frac{d}{dx}\mathbb{E}|Y-x| = F(x) - (1-F(x)) = 0 \Rightarrow F(x^*) = 1/2$ (the median)
  - pinball $\rho_\tau(e) = (\tau - 1\{e<0\})e$: $F(x) - \tau = 0 \Rightarrow x^* = F^{-1}(\tau)$ (the τ-quantile)
- Notation of the pinball loss
  - $\tau \in (0,1)$: the quantile level; $1\{e<0\}$: 1 if the forecast is too high, 0 otherwise; F^{-1}: the quantile function
  - under-forecasts cost $\tau|e|$, over-forecasts $(1-\tau)|e|$: a high τ pushes the forecast up
- Consequence: announce the loss *before* asking for a forecast; a median forecast judged by RMSE is judged unfairly Gneiting (2011)

Optimal point forecasts under three loss functions



- Y log-normal with $\ln Y \sim N(0, 0.6^2)$; expected loss as a function of the point forecast x (simulation, 2×10^5 draws)

ATS_ch1_loss_functions

Interpreting the three minimisers

- Mean 1.20, median 1, 0.9-quantile 2.16; three different "best" forecasts of the same variable
  - for a right-skewed target (wages, loads, prices, volatility) the gap is large
- The pinball curve is flat near its minimum: quantile forecasts are hard to separate with few observations
- A forecaster who knows the loss reports the matching functional; ranking forecasts by another loss rewards the wrong behaviour

Bregman losses: all losses consistent for the mean (1/2)

- **Theorem** Savage (1971), Gneiting (2011): under regularity, L is consistent for the mean iff it is a **Bregman** loss
  - $L(x, y) = \phi(y) - \phi(x) - \phi'(x)(y - x)$
  - ϕ: a convex function, ϕ' its derivative; L is the gap between $\phi(y)$ and the tangent of ϕ at x, so $L \geq 0$ (up to terms in y only)
- Two members of the family
  - $\phi(x) = x^2$: the squared error $(y - x)^2$
  - $\phi(x) = -\ln x$: **QLIKE** $y/x - \ln(y/x) - 1$, for a variance forecast $x > 0$ and a variance proxy $y > 0$ (Chapter 8)
  - the conditional mean is optimal for every Bregman loss Banerjee et al. (2005); proof for the quantile in the Appendix

Bregman losses: all losses consistent for the mean (2/2)

- Proof sketch, with $\mu = \mathbb{E}Y$:
  - $\mathbb{E}L(x, Y) - \mathbb{E}L(\mu, Y) = \phi(\mu) - \phi(x) - \phi'(x)(\mu - x) \geq 0$
  - the linear term vanishes because $\mathbb{E}(Y - \mu) = 0$; the inequality is convexity of ϕ
- Correctly specified forecasts rank the same under every Bregman loss; misspecified ones need not Patton (2020)
  - Murphy diagrams Ehm et al. (2016) plot the score over the whole family: one forecast dominates only if its curve is lower everywhere
  - robust proxies for volatility: only some losses keep the ranking with a noisy proxy Patton (2011)

Asymmetric loss

- **LinEx** loss $L(e) = b[\exp(ae) - ae - 1]$, $e = y - x$: linear on one side, exponential on the other
  - $b > 0$: a scale; a: the asymmetry; $a > 0$ penalises under-forecasts ($e > 0$) exponentially, over-forecasts about linearly
  - optimal forecast $x^* = a^{-1} \ln \mathbb{E} e^{aY}$; with $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$ (conditional mean μ_t, variance σ_t^2): $x^* = \mu_t + a\sigma_t^2/2$ Christoffersen and Diebold (1997)
  - the optimal forecast is **biased** and its bias moves with the conditional variance
- Consequences for evaluation
  - a rational forecaster under asymmetric loss fails the MZ unbiasedness test (Section 4)
  - the pinball loss is the piecewise-linear member: the optimal forecast is a quantile
- Examples: a grid operator (under-forecasting load is costlier than over-forecasting), a central bank with asymmetric inflation costs, the capital buffer of a bank (Chapter 9)

Recap: Point forecasts and loss functions

- A loss defines the target functional: squared $\rightarrow$ mean, absolute $\rightarrow$ median, pinball $\rightarrow$ quantile
- Bregman losses are exactly those consistent for the mean; QLIKE is one of them
- Under asymmetric loss the optimal forecast is biased by design
- With misspecified models the ranking can depend on the loss: fix it in advance

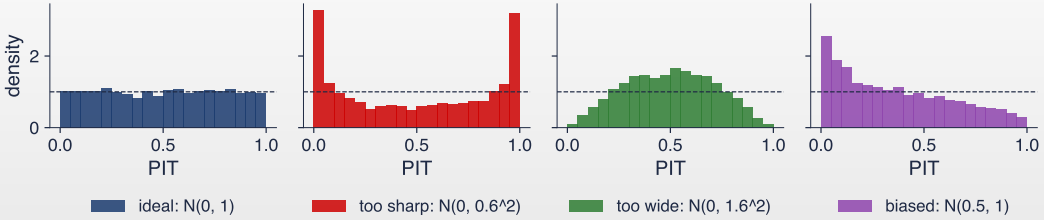
Maximise sharpness subject to calibration

- **Calibration**: statistical consistency between the forecasts F_t and the outcomes y_t (a joint property) Gneiting, Balabdaoui and Raftery (2007)
  - **probabilistic** calibration: the PIT (probability integral transform) $u_t = F_t(y_t)$, the forecast probability of an outcome below y_t, is uniform on average
  - **marginal** calibration: $\frac{1}{T}\sum_t F_t(x) \to G(x)$ for every x, with G the limit of the empirical distribution of y_t
- **Sharpness**: concentration of the predictive distributions; a property of the forecasts only
  - measured by the average width of central intervals or the average predictive variance
- Paradigm Gneiting, Balabdaoui and Raftery (2007): among calibrated forecasts, prefer the sharpest
  - the climatological forecast (the unconditional distribution) is calibrated but not sharp
  - proper scores (next section) reward both at once

The probability integral transform

- **Theorem** Dawid (1984), Diebold, Gunther and Tay (1998): if F_t is the true conditional distribution of Y_t given $\mathcal{F}_{t-1}$ and continuous, then $u_t = F_t(y_t)$ are i.i.d. $U(0,1)$
  - uniform: for every $v \in (0,1)$, $P(u_t \leq v \mid \mathcal{F}_{t-1}) = P(Y_t \leq F_t^{-1}(v) \mid \mathcal{F}_{t-1}) = v$, whatever the past
  - independent: a conditional distribution that does not depend on the past implies independence of the sequence
- For $h > 1$ the PITs are uniform but serially dependent up to lag $h-1$ (overlapping horizons)
  - tests must allow this dependence Rossi and Sekhposyan (2019)
- Uniformity is necessary, not sufficient: Hamill (2001) and Gneiting, Balabdaoui and Raftery (2007) build uncalibrated forecasters whose PIT is uniform on average
  - check PIT conditionally: by sub-sample, by regime, by horizon

PIT histogram shapes



- $Y \sim N(0, 1)$, 5000 draws; four Normal forecasts; dashed line: the uniform density

ATS_ch1_calibration_scores

Interpreting the PIT shapes

- **U-shape**: too many outcomes in both tails (32% outside the central 90%, against 10%): the forecast is too sharp, intervals too narrow
- **Hump**: too few outcomes in the tails (1.0%): the forecast is too wide, the intervals waste information
- **Slope**: a biased location; here the forecast mean is too high, so PIT values cluster near 0
- CRPS (lower is better): ideal 0.567, too sharp 0.595, too wide 0.604, biased 0.637: the proper score ranks the ideal forecast first

Testing calibration (1/2)

- **Berkowitz test** Berkowitz (2001): if the PITs are i.i.d. uniform, $z_t = \Phi^{-1}(u_t)$ is i.i.d. $N(0,1)$
  - Φ^{-1}: the quantile function of the standard Normal distribution
- Fit a Gaussian AR(1) to z_t and test its parameters:
  - $z_t = \mu + \rho(z_{t-1} - \mu) + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2(1-\rho^2))$
  - μ: the mean of z_t; σ^2: its variance; ρ: its first-order autocorrelation; calibration means $\mu = 0$, $\sigma = 1$, $\rho = 0$
  - LR (likelihood ratio) statistic $2(\ell_1 - \ell_0) \sim \chi^2(3)$ under H_0; ℓ_1, ℓ_0: maximised log-likelihoods without and with the restrictions
  - the transformation to z makes the tails visible; the test checks only the first two moments and AR(1) dependence

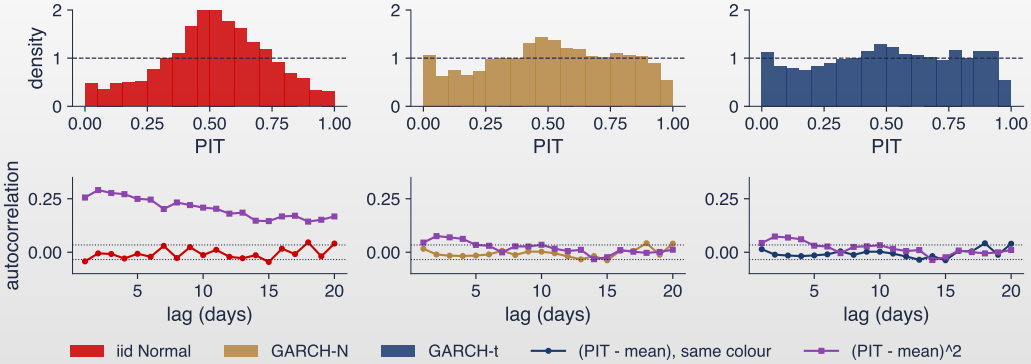
Testing calibration (2/2)

- **Correlograms** of $(u_t - \bar{u})^k$, $k = 1, 2$ Diebold, Gunther and Tay (1998): dependence in level and in volatility
  - $\bar{u}$: the mean PIT; Ljung–Box on $(u_t - \bar{u})^2$ detects missing volatility dynamics
- Parameter estimation changes the null distribution of PIT tests
  - with P/R small (few forecasts relative to the estimation sample) it can be ignored, otherwise use Rossi and Sekhposyan (2019)
- Interval forecasts: coverage and independence of the hits Christoffersen (1998)
  - hit: an outcome outside the interval; for VaR this becomes backtesting (Chapter 9)

Case study: Diebold, Gunther and Tay (1998)

- The paper: daily S&P 500 returns; parameters estimated once on the first part of the sample, density forecasts evaluated on the second part with PIT histograms and correlograms of $(u - \bar{u})^k$ Diebold, Gunther and Tay (1998)
  - their models: i.i.d. Normal; MA(1)-GARCH(1,1) with Normal and with Student-t innovations Bollerslev (1987)
- Our replication: the same three models, estimated by maximum likelihood on 2000–2012, fixed parameters, one-day density forecasts for 2013–2026 (3 448 days)
  - model: $r_t = c + \varepsilon_t + \theta\varepsilon_{t-1}$, $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; z_t Normal or Student-t with ν degrees of freedom
  - GARCH-t: $\hat{\theta} = -0.061$, $\hat{\alpha} + \hat{\beta} = 0.996$ (the persistence of volatility), $\hat{\nu} = 8.0$ (fat tails)
- Question: which density forecast is calibrated, and do the proper scores agree with the PIT diagnosis?

PIT diagnostics of three S&P 500 density forecasts



- Top: PIT histograms (20 bins); bottom: autocorrelations of $(u - \bar{u})$ and $(u - \bar{u})^2$ with $\pm 2/\sqrt{T}$ bands

ATS_ch1_density_forecasts

Interpreting the S&P 500 diagnostics

- i.i.d. Normal: a hump and strong autocorrelation of $(u - \bar{u})^2$: the forecast ignores volatility clustering; Berkowitz LR 428.8 ($p < 0.001$)
  - too wide in calm periods, too narrow in turbulent ones: the two errors average into a hump
- GARCH removes most of the dependence; the t version has the better left tail
  - Berkowitz: GARCH-N LR 9.3 (p 0.026), GARCH-t LR 10.0 (p 0.019); estimated σ_z 0.97 < 1
  - with parameters frozen in 2012 the forecasts are slightly too wide after 2013: fewer than 10% of days fall outside the central 90% (scores table, next section)
- In line with Diebold, Gunther and Tay (1998): conditional heteroskedasticity is essential, fat tails help; a small residual miscalibration remains

Recap: Calibration

- Calibration is a joint property of forecasts and outcomes; sharpness is a property of the forecasts
- Correct one-step densities give i.i.d. uniform PITs; for $h > 1$ they are dependent
- U-shape: too sharp; hump: too wide; slope: bias; correlogram of $(u - \bar{u})^2$: missing volatility dynamics
- The S&P 500: GARCH-t is close to calibrated, the i.i.d. Normal is not

Proper scoring rules

- A **scoring rule** $S(F, y)$ assigns a penalty to the forecast F when y occurs (negatively oriented: lower is better)
  - **proper**: $\mathbb{E}_G S(G, Y) \leq \mathbb{E}_G S(F, Y)$ for all F, G Gneiting and Raftery (2007)
  - G: the distribution the forecaster believes; $\mathbb{E}_G$: the expectation under G; reporting the true belief is optimal
  - **strictly proper**: equality only if $F = G$; then the score cannot be manipulated
- Proper scores reward calibration and sharpness together: the expected score decomposes into uncertainty, minus resolution, plus reliability
  - the Brier score $(p - 1\{\text{event}\})^2$, with p the forecast probability of a binary event, is the oldest example Brier (1950)
- A proper score of a probabilistic forecast reduces to a consistent loss of a point forecast when F is a point mass (example: the CRPS, two slides later)

The logarithmic score

- $\mathrm{LogS}(f, y) = -\ln f(y)$ Good (1952): the only proper score that is **local** (depends on f only at y, up to equivalence)
  - f: the forecast density; g: the true density; properness: $\mathbb{E}_g[\mathrm{LogS}(f, Y)] - \mathbb{E}_g[\mathrm{LogS}(g, Y)] = \int g \ln(g/f) = \mathrm{KL}(g \| f) \geq 0$
  - KL: the Kullback–Leibler divergence, nonnegative and zero only if $f = g$ (Gibbs inequality)
- Links: average log score = minus the out-of-sample log-likelihood; the difference of two log scores is a predictive likelihood ratio
  - Bayes factors and the predictive model choice of Geweke and Amisano (2011) are built on it
- Weaknesses: infinite if $f(y) = 0$; very sensitive to tail events; needs a density
  - cannot score ensembles or quantile forecasts directly

The continuous ranked probability score (1/2)

- Definition Matheson and Winkler (1976): the squared distance between the forecast distribution function and the step function of the outcome
  - $\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - 1\{y \leq z\})^2 dz$
  - z: a threshold; $1\{y \leq z\}$: the "distribution function" of the outcome, 0 below y and 1 above; units: those of y
- Kernel form Gneiting and Raftery (2007) (proof in the Appendix):
  - $\mathrm{CRPS}(F, y) = \mathbb{E}_F|X - y| - \frac{1}{2}\mathbb{E}_F|X - X'|, \quad X, X' \sim F$ independent
  - first term: accuracy; second: a reward for spread that stops the score from favouring point masses
  - for a point forecast (F a point mass at x) it reduces to the absolute error $|y - x|$

The continuous ranked probability score (2/2)

- Closed form for $F = N(\mu, \sigma^2)$, with $z = (y - \mu)/\sigma$:
  - $\mathrm{CRPS} = \sigma \left[z(2\Phi(z) - 1) + 2\varphi(z) - 1/\sqrt{\pi} \right]$
  - Φ, φ: the distribution and density functions of $N(0, 1)$; a closed form exists also for Student t (Quantlet)
  - ensembles: plug in the empirical distribution of the members
- Decompositions:
  - $\mathrm{CRPS} = \int \mathrm{BS}_z \, dz = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) \, d\tau$
  - BS_z: the Brier score of the event $\{y \leq z\}$; ρ_τ: the pinball loss at level τ
  - so the average pinball over the 99 percentiles (GEFCom2014) approximates CRPS/2 Hong et al. (2016)

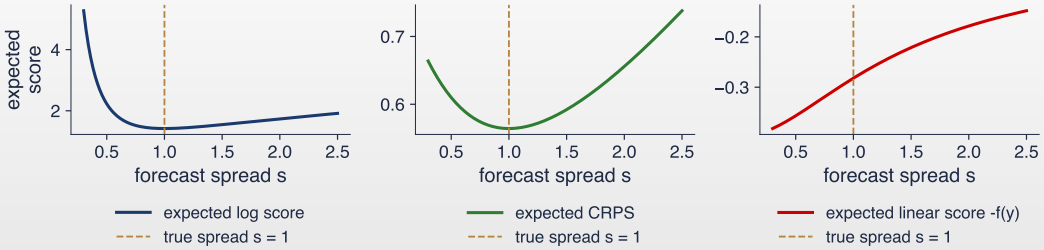
Quantile, interval and multivariate scores (1/2)

- **Quantile score**: $\rho_\tau(y - q)$ for a forecast q of the τ-quantile, strictly consistent Gneiting (2011)
  - any increasing g gives another consistent score: $(1\{y \le q\} - \tau)(g(q) - g(y))$
- **Interval score** of a central $(1 - \alpha)$ interval $[l, u]$ Gneiting and Raftery (2007)
  - $\mathrm{IS}_\alpha = (u - l) + \frac{2}{\alpha}(l - y)1\{y < l\} + \frac{2}{\alpha}(y - u)1\{y > u\}$
  - l, u: the lower and upper limits; width plus a penalty $2/\alpha$ per unit of miss
  - it equals $\frac{2}{\alpha}[\rho_{\alpha/2}(y - l) + \rho_{1-\alpha/2}(y - u)]$; coverage alone is not a score

Quantile, interval and multivariate scores (2/2)

- **Energy score** for vectors, the multivariate CRPS:
  - $\mathrm{ES}(F, \mathrm{y}) = \mathbb{E}\|\mathrm{X} - \mathrm{y}\| - \frac{1}{2}\mathbb{E}\|\mathrm{X} - \mathrm{X}'\|$
  - y: the outcome vector; X, X': independent draws from the forecast F; $\|\cdot\|$: the Euclidean norm
  - weak at detecting wrong correlations; the **variogram score** Scheuerer and Hamill (2015) targets the dependence structure (24 hourly loads, a VAR path)
  - $\mathrm{VS}_p = \sum_{i,j} w_{ij}\big(|y_i - y_j|^p - \mathbb{E}|X_i - X_j|^p\big)^2$: compares observed and forecast differences between components; $w_{ij} \geq 0$: weights, p: usually 0.5
- Tails: threshold-weighted CRPS stays proper Gneiting and Ranjan (2011)
  - $\int (F(z) - 1\{y \leq z\})^2 w(z) dz$
  - $w(z) \geq 0$: a weight that emphasises the thresholds of interest (e.g. the left tail); the weighted likelihood ratio of Amisano and Giacomini (2007) does not stay proper Gneiting and Ranjan (2011)

Proper and improper scores



- Truth $Y \sim N(0,1)$; forecast $N(0, s^2)$; expected score as a function of the spread s (closed forms)
- Log score and CRPS: minimum at $s = 1$; linear score $-f(y)$: $\mathbb{E}f(Y) = [2\pi(1+s^2)]^{-1/2}$ grows as $s \to 0$: it rewards overconfidence

ATS_ch1_calibration_scores

Interpreting the expected scores

- The log score and the CRPS are minimised by the true spread: a forecaster maximises the expected reward by reporting what she believes
- The log score punishes overconfidence much more than excess spread (steep left branch); the CRPS is more symmetric and bounded by the absolute error
- The linear score is maximised by a point mass: it would reward a forecaster who always reports a spike, so it must not be used for evaluation

Elicitability

- A functional T is **elicitable** if a strictly consistent loss exists for it Gneiting (2011)
  - mean, quantiles, expectiles, ratios of expectations: elicitable
  - only elicitable functionals can be compared and ranked meaningfully with point forecasts
- **Necessary condition** (Osband): convex level sets: $\mathrm{T}(F_0) = \mathrm{T}(F_1) = t \Rightarrow \mathrm{T}(\lambda F_0 + (1-\lambda)F_1) = t$
  - F_0, F_1: two distributions with the same value t; $\lambda \in [0,1]$: the mixing weight
  - the **variance** fails it, so it is not elicitable; the pair (mean, second moment) is
- Risk measures with $\mathrm{VaR}_\alpha(X) = -q_\alpha(X)$, q_α the α-quantile of the return X (VaR 1%: $\alpha = 0.01$)
  - VaR is a quantile: elicitable by the pinball loss
  - ES (expected shortfall, the mean loss beyond VaR) has non-convex level sets: **not elicitable** alone Weber (2006), Gneiting (2011); (VaR, ES) is **jointly** elicitable Fissler and Ziegel (2016)
  - the Fissler–Ziegel scores and the backtests built on them: Chapter 9

Scores of the S&P 500 density forecasts

Model	Log score	CRPS	Outside 90%	Berkowitz p
i.i.d. Normal	1.531	0.564	3.9%	< 0.001
MA(1)-GARCH(1,1)-N	1.265	0.509	8.0%	0.026
MA(1)-GARCH(1,1)-t	1.233	0.507	8.4%	0.019

- Score differences tested with DM on the daily scores (Amisano and Giacomini (2007), unweighted)
  - GARCH-t against GARCH-N: HLN -5.9 (log score), -6.4 (CRPS); GARCH-N against i.i.d.: -12.7 and -21.5
- Both scores rank the models in the same order as the PIT diagnosis; the log score separates the tails more sharply
- Ranking and calibration are different questions: the best-scoring model can still fail a calibration test

Recap: Proper scores and elicitability

- Proper scores make honesty optimal; strictly proper scores cannot be gamed
- LogS: local, tail-sensitive; CRPS: robust, works for ensembles, equals twice the integrated pinball loss
- Interval score for intervals, energy and variogram scores for vectors
- Elicitable: mean, quantile (VaR); not elicitable: variance, ES alone; (VaR, ES) jointly

Two questions about predictive ability

- **Population** West (1996): are the *models*, at their pseudo-true parameters, equally accurate?
  - estimation error enters the asymptotic variance unless $P/R \to 0$ or the loss is the one used in estimation
- **Finite sample** Giacomini and White (2006): are the *forecasting methods*, with their estimation windows, equally accurate?
  - rolling window of fixed size R: estimation noise is part of the method; nested models allowed
- **Diebold–Mariano** Diebold and Mariano (1995) treats the forecasts as given ("primitives"): it compares forecasts, not models Diebold (2015)
  - using DM to compare models with estimated parameters needs the West or GW framework

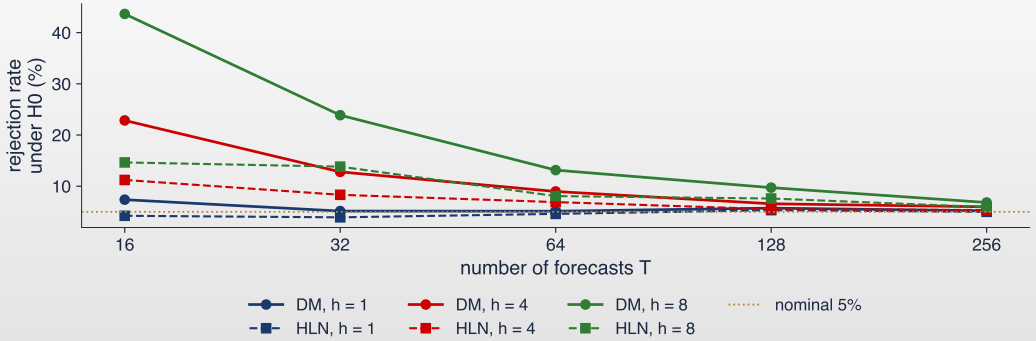
The Diebold–Mariano test with HAC variance (1/2)

- Loss differential $d_t = L(e_{1t}) - L(e_{2t})$; H_0: $\mathbb{E}d_t = 0$ (equal accuracy)
  - e_{it}: the error of forecast i at t; $d_t > 0$: forecast 2 did better at t; d_t assumed covariance stationary
- The statistic: the mean differential over its HAC standard error
  - $\mathrm{DM} = \bar{d}/\sqrt{\hat{\sigma}^2_{LR}/P} \to N(0,1), \quad \sigma^2_{LR} = \gamma_0 + 2\sum_{k\geq 1}\gamma_k$
  - $\bar{d}$: the mean of d_t over the P forecasts; γ_k: the autocovariances of d_t; σ^2_{LR}: its long-run variance (Chapter 0)
  - a large positive DM: forecast 2 is more accurate; reject at 5% if $|\mathrm{DM}| > 1.96$

The Diebold–Mariano test with HAC variance (2/2)

- Optimal h-step errors are at most MA($h-1$), so DM truncate at $h-1$ lags with a **rectangular** kernel
  - the rectangular estimate can be **negative**; Newey–West (Bartlett) weights keep it positive Newey and West (1987), at the cost of downward bias
  - suboptimal forecasts have errors with longer memory: use a data-driven bandwidth (Chapter 0)
- **HLN correction** Harvey, Leybourne and Newbold (1997): a small-sample rescaling of DM
  - $\mathrm{DM}^* = \mathrm{DM}\sqrt{(P+1-2h+h(h-1)/P)/P}$, compared with t_{P-1}
  - the factor (below 1) removes the bias of the variance estimator for MA($h-1$) errors; t_{P-1}: Student t with $P-1$ degrees of freedom, for small P

Size of DM and HLN in small samples



- Two independent MA($h-1$) error series with equal variance, squared loss; share of rejections at 5% in 4 000 replications

ATS_ch1_dm_tests

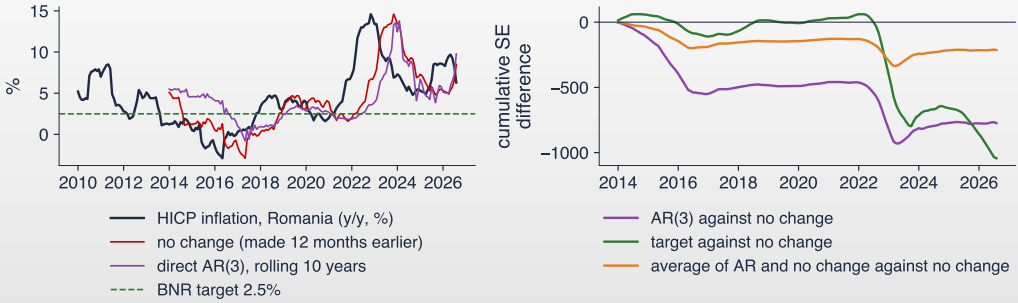
Interpreting the size experiment

- $h = 1$: both tests are close to 5%; DM over-rejects only at $P = 16$ (7.4%)
- $h = 8$, $P = 16$: DM rejects a true null 43.6% of the time, HLN 14.6%; at $P = 64$: 13.2% and 8.1%
  - the HAC variance with few observations and many lags is noisy and biased down
- Rule: at multi-step horizons with fewer than about 100 forecasts, use HLN and treat p-values near 5% as inconclusive; fixed-b critical values (Chapter 0) are an alternative

Romanian inflation one year ahead: the design

- Target: HICP inflation, y/y, 12 months ahead; forecast origins from January 2013, targets January 2014 to August 2026, $P = 152$ forecasts
  - HICP inflation in August 2026: 6.3%
- Forecasts
  - **no change**: $\hat{\pi}_{t+12} = \pi_t$; **direct AR(3)**: π_{t+12} on $1, \pi_t, \pi_{t-1}, \pi_{t-2}$, OLS on a 10-year rolling window
  - **target**: the BNR inflation target, 2.5%; **average** of AR(3) and no change
- Overlapping 12-month targets: errors are serially correlated up to lag 11; DM with $h = 12$

Romanian inflation forecasts one year ahead



- Left: inflation and the forecasts made 12 months earlier; right: cumulative sum of $e^2_{\mathrm{nc},t} - e^2_{m,t}$ (falling: model m loses to no change)

ATS_ch1_dm_tests

Interpreting the Romanian inflation comparison

Forecast	RMSE	DM	HLN	p (HLN)	p, abs. loss
no change	3.57	–	–	–	–
AR(3)	4.23	1.43	1.32	0.188	0.381
BNR target	4.43	1.30	1.21	0.230	0.319
average	3.76	0.96	0.88	0.378	0.795

- Every model has a higher RMSE than no change, yet no difference is significant: with $P = 152$ overlapping targets the effective sample is small
- Newey–West instead of the rectangular kernel: HLN 1.47 for AR(3); the conclusion does not depend on the kernel
- Relative RMSE of AR(3): 1.44 before July 2021, 1.10 after: the ranking is not stable over time Giacomini and Rossi (2010)

Conditional predictive ability: Giacomini–White (1/2)

- H_0: $\mathbb{E}[d_{t+h} \mid \mathcal{G}_t] = 0$ Giacomini and White (2006)
  - $\mathcal{G}_t$: the information at the forecast origin; H_0: no instrument $h_t \in \mathcal{G}_t$ predicts which method wins
  - h_t: a $q \times 1$ vector of instruments (not the horizon h), e.g. a constant and the last loss differential
- The statistic: a Wald test that the instruments do not predict d_{t+h}
  - $\mathrm{GW} = P\,\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \to \chi^2(q), \quad Z_t = h_t d_{t+h}$
  - $\bar{Z}$: the mean of Z_t; $\hat{\Omega}$: its HAC covariance matrix with $h-1$ lags; $\chi^2(q)$: chi-squared with q degrees of freedom
  - $h_t = 1$ gives the unconditional test; adding d_t or a state variable tests whether the past predicts the winner

Conditional predictive ability: Giacomini–White (2/2)

- Valid with a **rolling** window of fixed size, nested or non-nested models, any loss; invalid with a recursive window
  - a rejection suggests a decision rule: use model 1 when $\hat{\delta}' h_t > 0$, with $\hat{\delta}$ the regression of d_{t+h} on h_t
- Romanian inflation, AR(3) against no change, $h_t = (1, d_t, \pi_t - 2.5)$
  - $\pi_t - 2.5$: the distance of inflation from the BNR target
  - $\mathrm{GW} = 3.19$, p 0.363: neither the recent loss nor the distance from the target predicts the winner

Nested models: the failure of DM and the Clark–West test (1/2)

- Model 1 (small, e.g. the random walk) is nested in model 2; under H_0 the extra coefficients are zero
  - MSPE: the mean squared prediction error; population MSPEs are equal, and d_t is degenerate
  - so the DM statistic is not asymptotically Normal Clark and McCracken (2001)
- In finite samples model 2 estimates zero coefficients with noise
  - its MSPE is **larger** than that of model 1: DM is undersized (rejects too rarely) and has little power

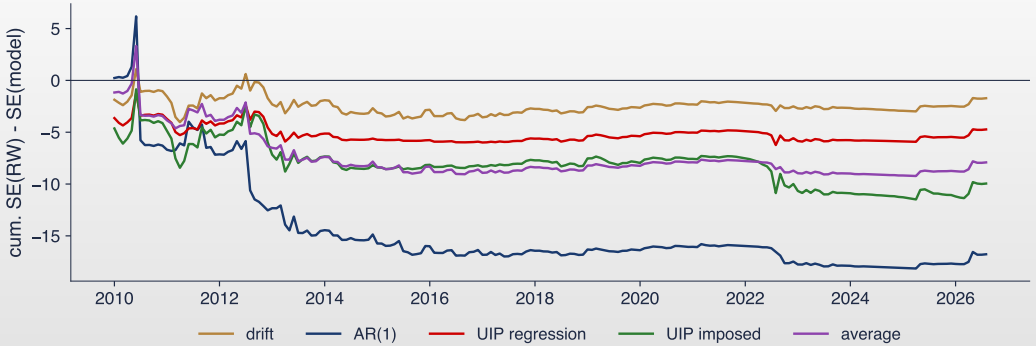
Nested models: the failure of DM and the Clark–West test (2/2)

- **Clark–West** Clark and West (2007): adjust the larger model for the noise it adds
  - $f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$
  - e_{1t}, e_{2t}: the errors of the small and large model; $\hat{y}_{1t}, \hat{y}_{2t}$: their forecasts; the squared gap between forecasts is the noise added by model 2
  - one-sided t test of $\mathbb{E}f_t = 0$ against $\mathbb{E}f_t > 0$ (model 2 better) with Normal critical values
- Why the adjustment works
  - under H_0, $\mathbb{E}e_1^2 - \mathbb{E}e_2^2 = -\mathbb{E}(\hat{y}_1 - \hat{y}_2)^2$: the adjustment recentres the differential (Appendix)
  - approximately Normal for rolling and recursive schemes; the exact critical values of Clark and McCracken (2001) are non-standard

Case study: Meese and Rogoff on EUR/RON

- Meese and Rogoff (1983): structural exchange-rate models do not beat the random walk out of sample at horizons up to one year
  - their design: rolling out-of-sample forecasts, RMSE against the random walk; the result has survived forty years
- Our data: monthly EUR/RON (BNR reference rate, end of month), $\Delta s_{t+1} = 100\Delta \ln S_{t+1}$; recursive estimation, forecasts January 2010 to August 2026 ($P = 200$)
  - S_t: the rate (lei per euro); Δs_{t+1}: its monthly log change in %, positive when the leu depreciates
  - nested in the random walk ($\Delta\hat{s} = 0$): drift, AR(p), the UIP regression $\Delta s_{t+1} = a + b(i_t^{RO} - i_t^{EA})/12 + u$ Fama (1984)
  - i_t^{RO}, i_t^{EA}: 3-month interest rates in % a year (divided by 12: per month); a, b: regression coefficients
  - no estimated parameters: UIP imposed ($a = 0, b = 1$), momentum (mean of the last three changes)
- UIP: uncovered interest parity, the expected depreciation equals the interest differential

EUR/RON: models against the random walk



- Cumulative $\sum_t (e^2_{RW,t} - e^2_{m,t})$ Welch and Goyal (2008): a line below zero means model m has accumulated more squared error than the random walk

ATS_ch1_nested_spa

Interpreting the EUR/RON tests

Model	RMSE / RW	DM (HLN)	CW	p (CW, one-sided)
drift	1.006	0.27	1.06	0.145
AR(1)	1.055	1.13	−0.15	0.561
AR(4)	1.062	1.29	−0.43	0.666
UIP regression	1.016	0.76	0.10	0.458
UIP imposed	1.033	0.99	–	–
momentum	1.183	2.61	–	–
average	1.026	0.85	–	–

- Every model has a higher RMSE than the random walk (random-walk RMSE 0.86 pp a month); the Meese–Rogoff result holds for the leu
- Clark–West does not reject either: the adjustment is not enough to make AR or UIP useful; only the drift gets close
- Managed float: the BNR smooths EUR/RON, so monthly changes are small and close to unpredictable from these predictors

Rationality: Mincer–Zarnowitz and encompassing (1/2)

- **Mincer–Zarnowitz** Mincer and Zarnowitz (1969): regress the outcome on the forecast
  - $y_{t+h} = a + b\hat{y}_{t+h|t} + u_{t+h}$
  - under squared loss an optimal forecast has $(a, b) = (0, 1)$: no bias, and a one-unit forecast change predicts a one-unit outcome change
  - Wald test of $(a, b) = (0, 1)$ with HAC errors (u is MA($h-1$)); $b < 1$: forecasts too volatile; $b > 1$: too timid
  - extended version: add variables in $\mathcal{F}_t$; a significant coefficient means unused information
- MZ assumes squared loss: under asymmetric loss a rational forecast is biased (Section 1)

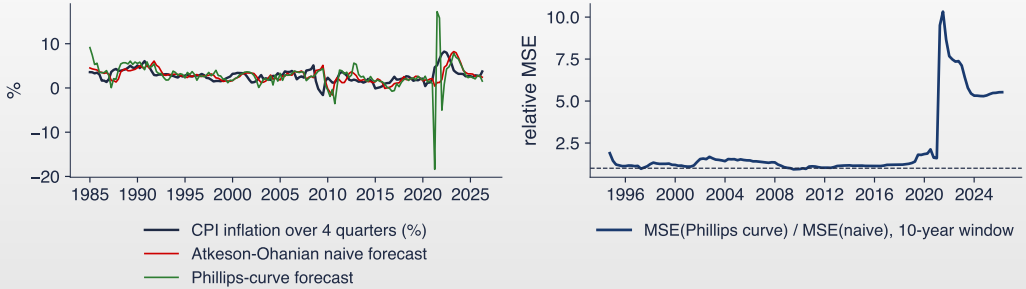
Rationality: Mincer–Zarnowitz and encompassing (2/2)

- **Encompassing** Chong and Hendry (1986), Harvey, Leybourne and Newbold (1998): forecast 1 encompasses forecast 2 if 2 adds nothing to 1
  - $y = (1-\lambda)\hat{y}_1 + \lambda\hat{y}_2 + \varepsilon, \quad H_0$: $\lambda = 0$
  - λ: the weight of forecast 2 in the best combination; $\lambda = 0$: forecast 2 is useless given forecast 1
  - HLN test: DM–HLN on $d_t = e_{1t}(e_{1t} - e_{2t})$, one-sided ($\mathbb{E}d_t > 0$ means $\lambda > 0$)
- If neither encompasses the other, a combination beats both (Section 5)

Case study: Atkeson and Ohanian (2001)

- Atkeson and Ohanian (2001): a naive forecast, "inflation over the next four quarters equals inflation over the last four quarters", beats Phillips-curve forecasts in the US after 1984
  - the Phillips curve of Stock and Watson (1999): $\pi^4_{t+4} - \pi^4_t = a + \beta(L)u_t + \gamma(L)\Delta\pi_t + e_{t+4}$
  - π^4_t: inflation over the last four quarters; π_t: quarterly inflation; u_t: the unemployment rate; $\beta(L), \gamma(L)$: lag polynomials (four lags each)
- Our update: CPI (quarterly average) and the unemployment rate (FRED); four lags of each, rolling window of 15 years; targets 1985Q1 to 2026Q2 ($P = 166$)
  - October 2025 CPI was not published; the 2025Q4 average uses two months
- Tests: DM–HLN with $h = 4$; GW with $h_t = (1, d_t)$ and $h_t = (1, u_t)$; encompassing in both directions

The naive forecast against the Phillips curve



- Left: four-quarter CPI inflation and the two forecasts made four quarters earlier; right: MSE ratio on a moving 10-year window

ATS_ch1_dm_tests

Interpreting the Atkeson–Ohanian update

- RMSE: naive 1.69, Phillips curve 2.97 (ratio 1.75); by period: 1.29 (1985–2007), 1.10 (2008–2019), 2.38 (2020–2026)
- Yet DM–HLN = 1.16, p 0.248: the large post-2020 errors inflate the HAC variance as much as the mean
  - the moving ratio peaks at 10.3 in 2021Q3: the Phillips curve extrapolated the pandemic unemployment spike
- GW: p-value 0.355 with the lagged differential, 0.322 with unemployment
  - encompassing, H_0 "the naive forecast encompasses the Phillips curve": p-value 0.289
  - encompassing, H_0 "the Phillips curve encompasses the naive forecast": p-value 0.118
- The AO conclusion holds in point estimates, and a test on 41 years of overlapping data cannot separate the two

Many models: data snooping, the reality check and SPA

- Testing k models against a benchmark and reporting the best p-value is **data snooping**
  - H_0: $\max_{j \le k} \mathbb{E}[d_{j,t}] \le 0$ with $d_{j,t} = L_{0,t} - L_{j,t}$: no model beats the benchmark
  - $L_{0,t}$: the loss of the benchmark; $L_{j,t}$: the loss of model j; $d_{j,t} > 0$: model j did better at t
- **Reality check** White (2000): statistic $\max_j \sqrt{P}\bar{d}_j$, p-value from the stationary bootstrap (Chapter 0) under the least favourable configuration
  - conservative: very poor models inflate the null distribution
- **SPA** Hansen (2005): studentise each $\bar{d}_j$ and drop clearly inferior models from the null distribution (consistent p-value)
  - lower and upper p-values bracket it; the upper one equals the RC with studentisation
  - StepM Romano and Wolf (2005) identifies *which* models beat the benchmark
- EUR/RON, $k = 9$ models (the seven of the table, AR(2) and AR(3)) against the random walk: SPA $p = 0.53$ (lower), 0.85 (consistent), 0.88 (upper): no model beats it

The model confidence set (1/2)

- **MCS** Hansen, Lunde and Nason (2011): the set $\hat{\mathcal{M}}_{1-\alpha}$ that contains the best model(s) with probability at least $1-\alpha$ asymptotically
  - no benchmark: all models are treated symmetrically; analogous to a confidence interval for a parameter
- Equivalence test on the m models still in the set
  - H_0: $\mathbb{E}[d_{i\cdot,t}] = 0$ for all i, $\quad d_{i\cdot,t} = L_{i,t} - \frac{1}{m}\sum_j L_{j,t}$
  - $d_{i\cdot,t}$: the loss of model i minus the average loss of all models; positive: model i is worse than average
  - statistic $T_{\max} = \max_i t_i$, $t_i = \bar{d}_{i\cdot}/\widehat{\mathrm{se}}(\bar{d}_{i\cdot})$: the worst model relative to the average, standardised

The model confidence set (2/2)

- Algorithm: start with all models
  - if the equivalence test rejects, **eliminate** the model with the largest t_i and repeat; stop at the first non-rejection
  - a block bootstrap gives the null distribution and the standard errors
  - MCS *p*-value of a model: the largest test *p*-value up to its elimination; it stays in $\hat{\mathcal{M}}_{1-\alpha}$ if this is above α
- A large MCS is a finding too
  - the data cannot separate the models (a weak sample, not equal models)

Case study: day-ahead electricity load in Romania

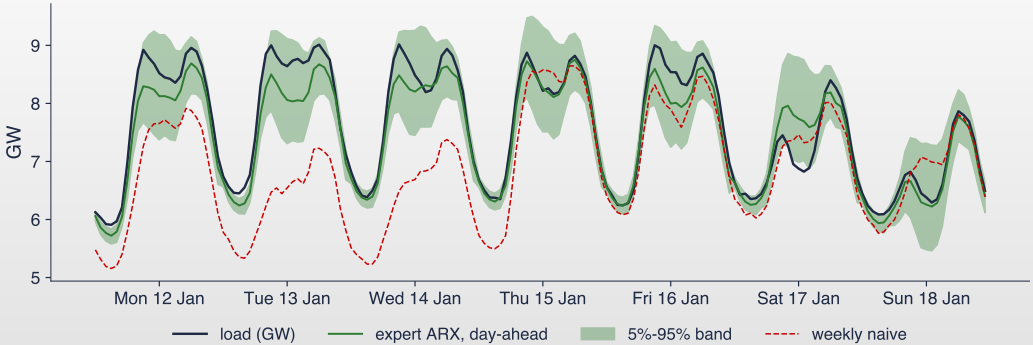


A 400 kV line, Predeluș Pass (2018)

Photo: Drprv (2018); CC BY-SA 3.0; Wikimedia Commons

- Target: the 24 hourly loads of day $d+1$ with data up to day d; 1 January 2025 to 30 September 2026 (638 days); mean 5.98 GW
  - quantile forecasts at the 99 percentiles, scored by the average pinball loss as in GEFCom2014 Hong et al. (2016), Hong and Fan (2016)
- Models: naive day, weekly naive, mean of 4 weeks; the **expert ARX** of Ziel and Weron (2018) for each hour
  - regressors: $y_{d,h}$, $y_{d-\mathbf{1},h}$, $y_{d-\mathbf{6},h}$, min and max of day d, last hour of d, Monday, Saturday, Sunday; one-year rolling window
  - quantiles: point forecast plus the empirical quantiles of the last 182 errors at the same hour
- ARX: autoregression with exogenous variables (here calendar dummies)

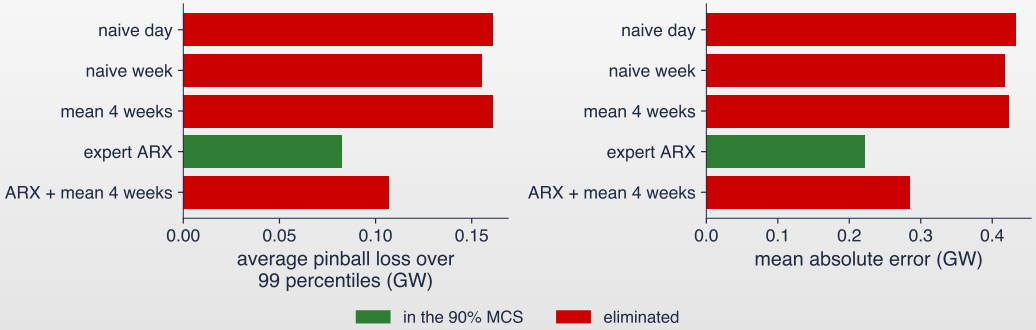
A winter week of load forecasts



- 12–18 January 2026; the weekly naive forecast copies the holiday week of 5–9 January (Orthodox Epiphany and St John, 6–7 January)

ATS_ch1_load_mcs

Average pinball loss and the 90% MCS



- Daily average losses; MCS with $T_{\max}$, moving-block bootstrap (blocks of 7 days), 2000 replications; green: in the MCS

ATS_ch1_load_mcs

Interpreting the load comparison

Model	Pinball (MW)	MAE (MW)	90% coverage	MCS p
naive (day d)	161	433	89.2%	< 0.001
weekly naive	155	417	87.6%	< 0.001
mean of 4 weeks	161	423	88.7%	< 0.001
expert ARX	83	221	88.2%	1.000
ARX + mean of 4 weeks	107	285	88.3%	< 0.001

- The expert ARX halves the losses of the naive rules and is alone in the MCS; DM–HLN against the ARX + mean combination: -10.7
- Here combining with a much worse forecast hurts: equal weights help only when the forecasts are of similar quality (Section 5)
- All intervals under-cover slightly; the largest losses fall on and right after public holidays (8 January 2026, 25 December 2025, 1 January 2025): calendar effects are the next improvement

Recap: Comparing forecasts

- DM compares forecasts; with estimated models choose West (population) or GW (methods, rolling window)
- Multi-step, small P: HAC variance plus the HLN correction and t_{P-1}
- Nested models: Clark–West; many models: SPA or the MCS, never the best p-value
- Romanian inflation, US inflation, EUR/RON: the simple benchmark is hard to beat and differences are rarely significant

Bates and Granger (1969)



Clive W. J. Granger (1934–2009), Nobel Memorial Prize 2003

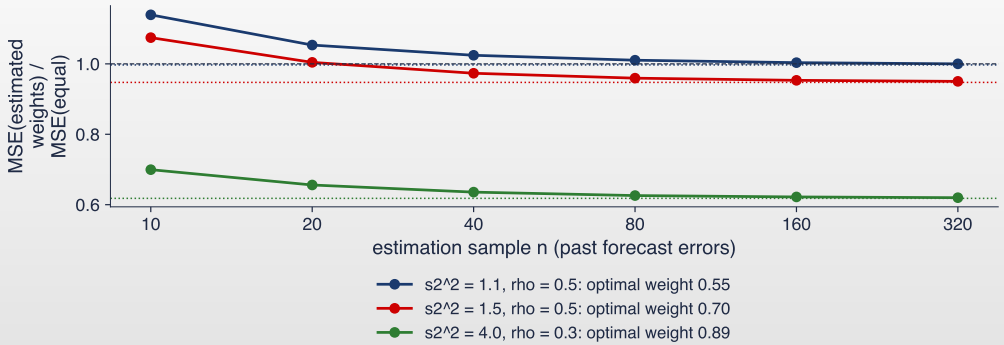
Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- Two unbiased forecasts with error variances σ_1^2, σ_2^2 and covariance σ_{12}; combination $w\hat{y}_1 + (1-w)\hat{y}_2$ Bates and Granger (1969)
  - error variance $w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\sigma_{12}$, minimised at
  - $w^* = \dfrac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \quad \sigma_c^2 = \dfrac{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \leq \min(\sigma_1^2, \sigma_2^2)$
  - σ_c^2: the error variance of the optimal combination, never above that of the better forecast; the less precise forecast gets the smaller weight
- N forecasts: $w^* = \Sigma^{-1}\iota/(\iota'\Sigma^{-1}\iota)$, with Σ the covariance matrix of the errors and ι a vector of ones; Granger and Ramanathan (1984): regress y on the forecasts, with or without a constant and the sum-to-one restriction
  - combination diversifies model risk, structural breaks and the private information of forecasters Timmermann (2006)

The forecast combination puzzle

- Empirical regularity: the simple average beats the "optimal" estimated weights Stock and Watson (2004), Timmermann (2006), Genre et al. (2013)
  - Stock–Watson: across 7 countries and many predictors of output growth, equal weights and the median are among the best combinations
- **Explanation** Smith and Wallis (2009), Claeskens et al. (2016): the weights are estimated
  - $\mathbb{E}[\mathrm{MSE}(\hat{w})] = \mathrm{MSE}(w^*) + \mathbb{E}(\hat{w} - w^*)^2 D$, $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$ (the variance of $e_1 - e_2$)
  - $\hat{w}$: the weight estimated from n past errors; MSE: the mean squared error of the combination
  - gain of w^* over $1/2$: $(w^* - 1/2)^2 D$; loss from estimating: $\mathrm{Var}(\hat{w})D$, of order $1/n$
  - equal weights win when $(w^* - 1/2)^2 < \mathrm{Var}(\hat{w})$: similar forecasts, short samples, correlated errors
- Also: weights move with breaks, so long windows are biased and short ones noisy

The puzzle by simulation



- Bivariate Normal errors with $\sigma_1^2 = 1$; weights estimated from n past errors; 4 000 replications; dotted: population ratio $\mathrm{MSE}(w^*)/\mathrm{MSE}(1/2)$

ATS_ch1_combination

Interpreting the puzzle simulation

- Similar forecasts ($w^* = 0.55$): the optimal weight lowers the population MSE only to 0.997 of the equal-weight MSE, while estimating it costs: ratio 1.14 at $n = 10$, 1.024 at $n = 40$
- Moderately different ($w^* = 0.70$): estimated weights need about 20 past errors to break even (1.07 at $n = 10$, 0.973 at $n = 40$)
- Very different ($w^* = 0.89$): estimate the weights, or drop the weak forecast; the load case of Section 4 is of this kind

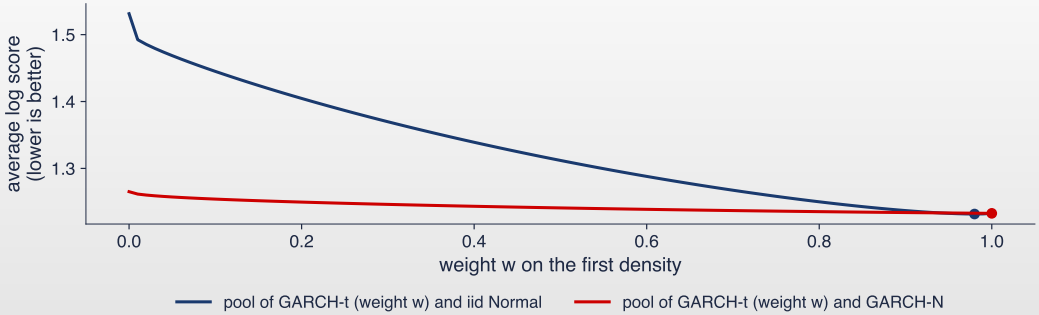
Between equal and optimal: shrinkage and robust combinations

- **Shrinkage**: $\hat{w}_\lambda = \lambda\hat{w} + (1-\lambda)\iota/N$ Diebold and Pauly (1990), Stock and Watson (2004)
  - $\hat{w}$: the estimated weights; ι/N: equal weights for N forecasts; $\lambda \in [0,1]$: $\lambda = 0$ gives the mean, $\lambda = 1$ the estimated weights
  - a Bayesian prior centred on equal weights; λ falls with the number of forecasts and rises with the sample
- **Performance weights**: $w_i \propto 1/\mathrm{MSE}_i$ on a recent window, ignoring correlations; discounting gives recent errors more weight
  - **previous best**: select the forecaster with the lowest past MSE (a weight of one)
- **Robust**: median, trimmed mean: protect against outlying forecasters
- Panels of experts change: forecasters enter and exit, so Σ is never fully observed Capistrán and Timmermann (2009)
  - performance weights need a minimum record; a review of 50 years of practice: Wang et al. (2023)

Combining density forecasts

- **Linear pool**: $p(y) = \sum_i w_i p_i(y)$, $w_i \geq 0$, $\sum w_i = 1$ Hall and Mitchell (2007)
  - p_i: the density forecast of model i; w_i: its weight; the pool is a mixture of the densities
  - weights that minimise the average log score: **optimal prediction pools** Geweke and Amisano (2011)
  - interior weights even when every model is misspecified: a pool is not a model average in the Bayesian sense
- The linear pool of calibrated densities is **over-dispersed** Gneiting and Ranjan (2013)
  - the mixture adds the variance of the means; remedy: a beta-transformed pool or quantile averaging (Vincentisation)
- Proper scores (Section 3) are the natural criteria for estimating and comparing pools

Optimal pools for the S&P 500 densities



- Average log score of $wp_{\text{GARCH-}t} + (1-w)p_{\mathbf{2}}$ on 2013–2026; dots: minima
- With the i.i.d. Normal: $w^* = 0.98$, score 1.2318 against 1.2328 for GARCH-t alone; with GARCH-N: $w^* = 1.00$: GARCH-t dominates, the pool adds almost nothing

ATS_ch1_density_forecasts

Interpreting the optimal pools

- The pool with the i.i.d. Normal puts almost all weight on GARCH-t: a small Normal component only insures against extreme days
- With GARCH-N the optimum is the corner $w = 1$: the two GARCH densities carry the same information, the t tails are simply better
- Weights estimated on the evaluation sample are optimistic; in practice estimate them on a past window and evaluate on the next

Case study: combination schemes against the simple average

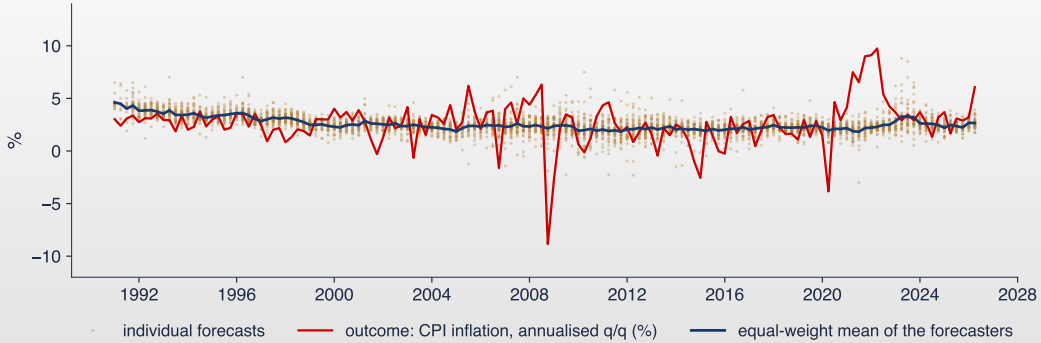


Federal Reserve Bank of Philadelphia, home of the SPF since 1990

Photo: Beyond My Ken (2013); CC BY-SA 4.0; Wikimedia Commons

- Genre et al. (2013) compare combination schemes on the ECB Survey of Professional Forecasters: equal weights, median, trimmed mean, performance-based weights, shrinkage, and others
  - their answer: rarely, and not robustly over time
- We apply their schemes to the US SPF: CPI inflation (annualised q/q) four quarters after the survey quarter; surveys 1990Q1–2025Q2 ($P = 142$, 195 forecasters, on average 36 per survey)
  - performance weights: inverse MSE over the last 20 surveys with a known outcome, at least 8 past forecasts (on average 25 eligible); trimmed mean: 10% on each side

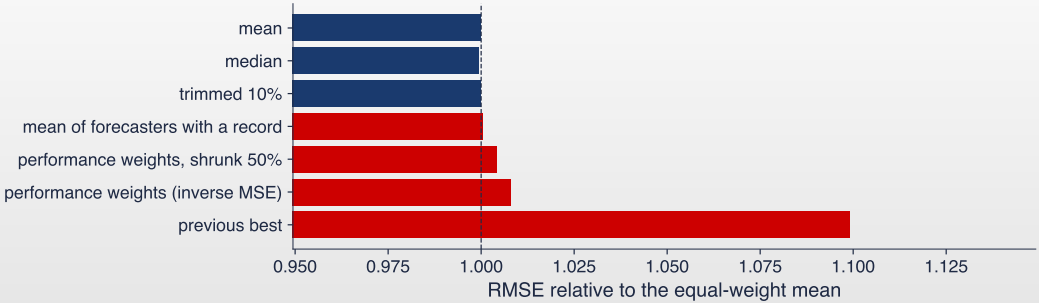
Individual SPF forecasts and the outcome



- Each dot: one forecaster's CPI forecast for the quarter four quarters ahead, placed at the target quarter; the outcome is far more volatile than any forecast

ATS_ch1_spf_combination

Combination schemes against the mean



- RMSE relative to the equal-weight mean (RMSE 2.18 pp); blue: at least as good as the mean

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Interpreting the SPF combination

- Median 0.999, trimmed mean 1.000, inverse-MSE 1.008, previous best 1.099 (relative RMSE)
  - DM–HLN against the mean: median p 0.842; inverse MSE p 0.091; previous best HLN 2.76, p 0.007
  - The answer of Genre et al. (2013) holds on US data: nothing beats the average significantly, and chasing the past winner loses
- Yet 33% of the 76 forecasters with at least 20 forecasts have a lower RMSE than the mean
  - they are known only ex post, over different and shorter samples: survivorship and luck, not skill one could have selected
- Mincer–Zarnowitz on the mean: $\hat{b} = 0.24$ (SE 0.31), $R^2 = 0.005$, $p < 0.001$: one-year-ahead quarterly CPI inflation is almost unpredictable; the forecasts are smooth, the outcomes noisy

Results of the M4 and M5 competitions

- **M4** (2018) Makridakis et al. (2020): 100 000 series, 61 methods, point forecasts and 95% intervals
  - the winner was a hybrid of exponential smoothing and a recurrent neural network; the second, a combination of statistical methods with weights learned by a meta-learner
  - combinations dominated the top places; pure machine learning methods did poorly; prediction intervals were too narrow for most methods
- **M5** (2020) Makridakis et al. (2022): 42 840 hierarchical Walmart sales series, weighted scaled errors
  - gradient-boosted trees (LightGBM) trained *across* series won; exogenous variables (prices, events) and cross-learning mattered
  - again, ensembles of many models were at the top (Chapter 12)
- Common lesson: combine, evaluate with scale-free scores (MASE, Chapter 0 of TSA) and proper scores, and report uncertainty that is calibrated

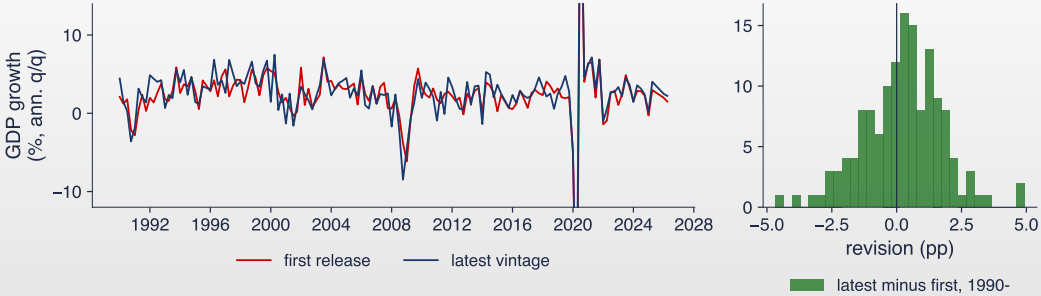
Recap: Combination

- Optimal weights reduce the error variance; estimated weights add a cost of order $1/n$
- The puzzle: when forecasts are similar, equal weights, the median and the trimmed mean win
- Linear pools of densities are over-dispersed; estimate them by the log score
- US SPF: no scheme beats the mean significantly; selecting the past winner loses

Vintages and revisions

- Macroeconomic data are revised: US GDP is first published about 30 days after the quarter and revised for years (benchmark revisions, new methods)
  - a **vintage** is the data set as available on a given date; the real-time data set stores all vintages Croushore and Stark (2001)
- Sources without a key: ALFRED (St. Louis Fed, vintages of FRED series), the Philadelphia Fed RTDSM; the ECB keeps a real-time database for the euro area; INS publishes a flash estimate of GDP and later revised estimates
  - forecasts made with latest-vintage data overstate what was predictable in real time Stark and Croushore (2002), Croushore (2011)
- Question for the evaluation: which vintage is the "truth"? The first release (what forecasters target) or the latest (the best estimate of reality)?

US real GDP growth: first release and latest vintage



- Annualised q/q growth, 1990–2026; revisions = latest minus first release (Philadelphia Fed RTDSM); the 2025Q4 first release is missing (government shutdown)

ATS_ch1_real_time

Interpreting the revisions

- Mean revision 0.23 pp, standard deviation 1.54 pp, mean absolute revision 1.21 pp over 143 quarters; correlation first–latest 0.939
  - largest: 2020Q2 (4.9 pp) and 2008Q4 (-4.7 pp): revisions are large exactly at turning points
- SPF current-quarter nowcast ($P = 143$): RMSE 1.95 against the first release, 2.36 against the latest; MZ slope 1.15 and 1.08
- The ranking of forecasters and models can change with the vintage: state the choice in the pre-registration

A pre-registered evaluation design

- Fix **before** looking at the out-of-sample results
  - target, horizons, loss or score (Sections 1–3), vintage of the outcome, evaluation period, estimation scheme and window
  - benchmark (random walk, seasonal naive, survey mean) and the full list of competing models
  - tests: DM–HLN or GW; Clark–West if nested; SPA or MCS for many models; a correction for several horizons (Holm)
- Report everything that was tried; a model selected on the test period has a biased error
  - the project of this course starts with exactly such a plan (proposal and pre-registration, 5%)

Recap: Real-time data

- Evaluate forecasts with the data that existed when they were made
- Revisions are large at turning points; the choice of the "truth" vintage changes RMSE and rankings
- Pre-register targets, losses, benchmarks, tests and the vintage

An open question

- Does the forecast combination puzzle survive when the panel of experts changes over time and the outcome distribution shifts (2020–2023)?
  - formal: H_0: $\mathbb{E}[(y - \hat{y}_\lambda)^2 - (y - \bar{y})^2] \geq 0$ for every shrinkage factor λ fixed in advance, in every sub-period
  - $\hat{y}_\lambda$: the shrinkage combination with factor λ; $\bar{y}$: the mean of all forecasters
  - falsified by a pre-registered λ that beats the mean with DM–HLN $p < 0.05$ after a Holm correction across sub-periods
- Why it matters: central banks publish the survey mean; a better combination would change the published expectations
  - literature to start from: Genre et al. (2013), Capistrán and Timmermann (2009), Smith and Wallis (2009), Claeskens et al. (2016), Wang et al. (2023)

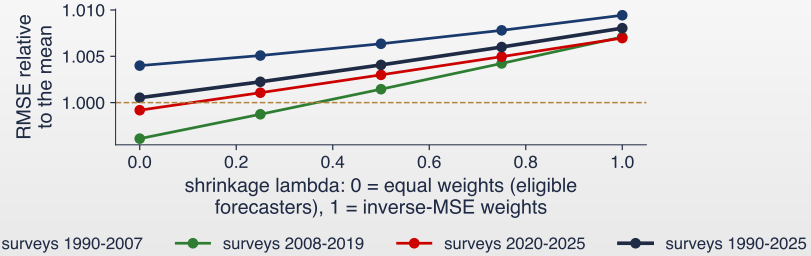
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers on combining SPF forecasts with entry and exit of forecasters; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Propose three reasons why inverse-MSE weights could beat the mean after 2020, each with a test.`
  - **code and replication**: ask for a function, then reproduce a known number first (the relative RMSE of the median in this lecture)
  - **robustness and critique**: `Act as a hostile referee: list every way this evaluation could be data-snooped.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- No look-ahead: weights at survey s use only outcomes known at s (here, forecasts from at least five surveys earlier, whose target quarter has ended)
- The HAC lag matches the overlap of the targets ($h - 1$); the HLN correction is applied
- λ, sub-periods and the minimum record are fixed before the results; all variants are reported
- The interpretation separates "not significant" from "equal"

Mini-case: shrinkage across sub-periods



- RMSE of λ·(inverse-MSE weights) + $(1-\lambda)$·(equal weights among eligible forecasters), relative to the mean of all forecasters; US SPF, CPI four quarters ahead
- $\lambda = 1$: 1.009 (1990–2007, $P = 72$), 1.007 (2008–2019), 1.007 (2020–2025); full sample HLN 1.70, p 0.091
- Every sub-period: the more weight on past performance, the worse; the puzzle survives the 2020–2023 inflation surge

ATS_ch1_spf_combination

Project idea

- **Combining Romanian inflation forecasts**: models (AR, Phillips curve with the output gap, BNR target), surveys and market-based expectations
  - pre-register horizons 1–12 months, the pinball loss at 5%, 50%, 95% and the CRPS, the Holm correction across horizons
  - replicate first: the Romanian inflation table of this lecture (Seminar 1, C1 does it by horizon)
  - extension: density combination by optimal pools; GW test of whether the 2022 shock changed the winner
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Choose the loss or score first: it defines what a "good" forecast is (mean, median, quantile, distribution)
- Density forecasts: check calibration with PIT, rank with proper scores; ES needs VaR to be scored
- Tests: DM–HLN with HAC, GW for methods, Clark–West for nested models, SPA and MCS for many
- Combine similar forecasts with equal weights; estimated weights pay off only with very different forecasts and long records
- Evaluate in real time and pre-register the design

Self-assessment

Questions

- Which functional does the pinball loss with $\tau = 0.05$ elicit?
- What does a U-shaped PIT histogram say about the intervals?
- Why is DM not valid for nested models with estimated parameters?
- When do estimated combination weights beat equal weights?
- Why can ES not be ranked on its own?

Next:

- Structural breaks and nonlinear models: tests for breaks, threshold models
- Forecast evaluation under instability: rolling windows, forecast breakdowns, the fluctuation test Giacomini and Rossi (2010)

Appendix: consistency of the pinball loss

- $\mathbb{E}\rho_\tau(Y - x) = \tau \int_x^\infty (y - x) dF(y) + (1 - \tau) \int_{-\infty}^x (x - y) dF(y)$
- Derivative (Leibniz): $-\tau(1 - F(x)) + (1 - \tau)F(x) = F(x) - \tau$; zero at $x = F^{-1}(\tau)$; second derivative $f(x) \geq 0$
- For $\tau = 1/2$: $\rho_{1/2}(e) = |e|/2$, so the median; for x off the quantile the excess loss is $\int_q^x (F(z) - \tau) dz > 0$
- Generalised piecewise linear losses $(1\{y \leq x\} - \tau)(g(x) - g(y))$, g increasing, are the whole class of consistent losses for the quantile Gneiting (2011)

Back

Appendix: the kernel form of the CRPS

- Write $(F(z) - 1\{y \leq z\})^2 = F(z)^2 - 2F(z)1\{y \leq z\} + 1\{y \leq z\}$
- With $X, X' \sim F$ i.i.d.: $F(z)^2 = P(X \leq z, X' \leq z)$ and $\mathbb{E}|X - y| = \int [F(z)(1 - 1\{y \leq z\}) + (1 - F(z))1\{y \leq z\}]dz$
- $\frac{1}{2}\mathbb{E}|X - X'| = \int F(z)(1 - F(z))dz$; subtracting gives $\int (F(z) - 1\{y \leq z\})^2 dz$
- Propriety: $\mathbb{E}_G \mathrm{CRPS}(F, Y) - \mathbb{E}_G \mathrm{CRPS}(G, Y) = \int (F(z) - G(z))^2 dz \geq 0$, zero only if $F = G$

Appendix: the Clark–West adjustment

- Under H_0 the larger model's extra coefficients $\beta_2 = 0$; its forecast is $\hat{y}_2 = \hat{y}_1 + \hat{\beta}_2' x_2$ approximately, with $\hat{\beta}_2$ pure estimation noise
- $e_2 = e_1 - (\hat{y}_2 - \hat{y}_1)$, so $e_1^2 - e_2^2 = 2e_1(\hat{y}_2 - \hat{y}_1) - (\hat{y}_2 - \hat{y}_1)^2$
- $\mathbb{E}[e_1(\hat{y}_2 - \hat{y}_1)] = 0$ under H_0 (e_1 is a martingale difference), hence $\mathbb{E}(e_1^2 - e_2^2) = -\mathbb{E}(\hat{y}_2 - \hat{y}_1)^2 < 0$
- Adding $(\hat{y}_1 - \hat{y}_2)^2$ recentres the differential at zero: $f_t = 2e_{1t}(\hat{y}_{2t} - \hat{y}_{1t})$, the encompassing statistic of Harvey, Leybourne and Newbold (1998) in disguise

Back

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